

Power BI Dashboard & Business Insights

Pizza Sales Analysis Dashboard

Microsoft Power BI

Introduction

This document explains the complete Power BI development process for the Pizza Sales Analysis project. It covers data preparation using Power Query, DAX measure creation, dashboard design, interactive features, and business insights derived from the dashboard.

The objective of this dashboard is to transform raw sales data into meaningful business insights that support better decision-making regarding sales performance, customer behavior, product performance, and operational planning.

Project Overview

Project Title

Pizza Sales Analysis Dashboard

Objective

To analyze pizza sales performance and answer important business questions such as:

- How much revenue did the business generate?
 - What is the average order value?
 - When do customers place the most orders?
 - Which pizza category performs the best?
 - Which pizza sizes contribute the highest revenue?
 - Which pizzas are the best and worst sellers?
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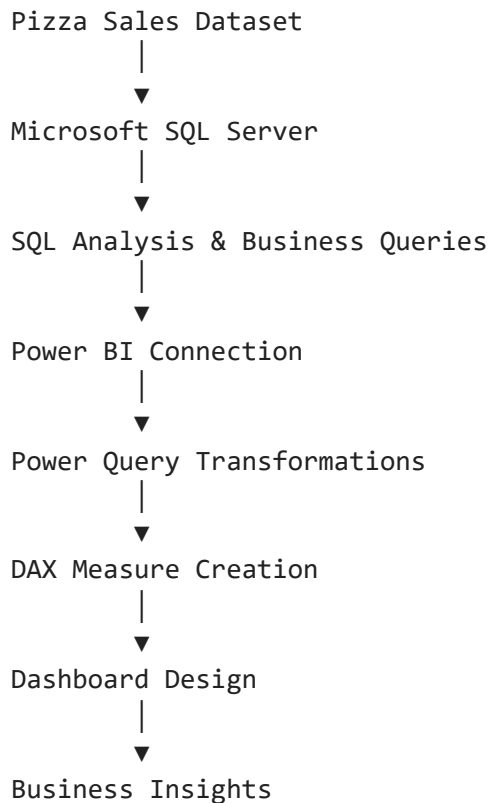
Tools & Technologies Used

Tool	Purpose
Microsoft SQL Server	Data Storage & Business Analysis
SQL Server Management Studio (SSMS)	SQL Query Execution
Microsoft Power BI Desktop	Dashboard Development
Power Query Editor	Data Cleaning & Transformation

	Tool	Purpose
	DAX	KPI Calculations & Measures

Data Import Process

The following workflow was followed while developing the dashboard.



Power Query Transformations

Power Query was used to clean and prepare the dataset before loading it into the Power BI model.

1. Standardized Pizza Size Values

The original dataset stored pizza sizes using abbreviated values.

Original Value	Updated Value
L	Large
M	Medium
R	Regular
XL	X-Large
XXL	XX-Large

Purpose

- Improve readability

- Create meaningful slicers
 - Display user-friendly chart legends
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2. Date Transformations

Additional columns were created from the original Order Date column.

Column Created	Purpose
Day Name	Daily Sales Analysis
Day Number	Correct Weekday Sorting
Month Name	Monthly Trend Analysis
Month Number	Correct Month Sorting

Why Day Number & Month Number?

Power BI sorts text alphabetically.

Without helper columns:

April
August
December
February
January

Using **Sort by Column**, the report correctly displays:

January
February
March
April
...
December

3. Time Transformation

A new **Hour** column was created from the **Order Time** column.

Purpose

Used to build the **Hourly Trend for Total Orders** visual.

Power BI Data Model

The dashboard uses a single fact table:

pizza_sales

The table contains:

- Order Information
- Pizza Details
- Quantity
- Revenue
- Date & Time
- Category
- Size

Since the dataset is relatively small, a single-table model is sufficient for this project.

DAX Measures

Several DAX measures were created to calculate KPIs dynamically.

Total Revenue

Total Revenue =
`SUM(pizza_sales[total_price])`

Purpose

Calculates total revenue generated from pizza sales.

Total Orders

Total Orders =
`DISTINCTCOUNT(pizza_sales[order_id])`

Purpose

Counts the total number of unique customer orders.

Total Pizzas Sold

Total Pizzas Sold =
`SUM(pizza_sales[quantity])`

Purpose

Calculates the total quantity of pizzas sold.

Average Order Value

Avg Order Value =
`DIVIDE([Total Revenue],[Total Orders],0)`

Purpose

Calculates the average amount spent by customers per order.

Average Pizzas Per Order

Avg Pizzas Per Order =
DIVIDE([Total Pizzas Sold],[Total Orders],0)

Purpose

Calculates the average number of pizzas purchased in each order.

Order Day

Order day =
UPPER(LEFT(pizza_sales[Day Name],3))

Purpose

Creates abbreviated weekday names for compact chart labels.

Order Month

Order Month =
UPPER(LEFT(pizza_sales[Month Name],3))

Purpose

Creates abbreviated month names for cleaner dashboard visuals.

Dashboard Features

The dashboard includes several interactive features to enhance user experience.

- ✔ KPI Cards
 - ✔ Date Range Filter
 - ✔ Pizza Category Slicer
 - ✔ Interactive Cross Filtering
 - ✔ Navigation Buttons
 - ✔ Multi-page Dashboard
 - ✔ Dynamic Charts
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Dashboard Pages

Page 1

Executive Dashboard

Contains:

- KPI Cards
- Hourly Trend
- Daily Trend
- Monthly Trend
- Sales by Category
- Sales by Size
- Total Quantity by Category

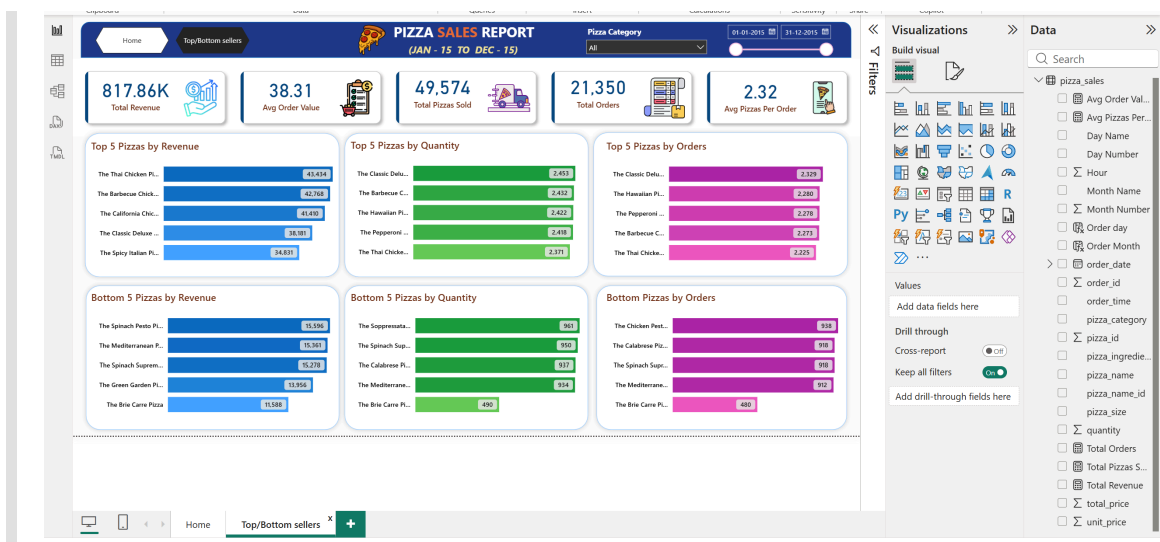


Page 2

Top & Bottom Sellers

Contains:

- Top 5 Revenue
- Bottom 5 Revenue
- Top 5 Quantity
- Bottom 5 Quantity
- Top 5 Orders
- Bottom 5 Orders



Dashboard Visualizations

Visual	Purpose
KPI Cards	Executive Summary
Line Chart	Monthly Trend
Area Chart	Hourly Orders
Column Chart	Daily Orders
Donut Chart	Sales Distribution
Horizontal Bar Chart	Product Comparison
Slicers	Interactive Filtering

Business Insights

Business Days & Time

Hourly Sales Trend

Insights

- Customer orders begin increasing significantly after **11:00 AM**.
- The **12:00 PM – 1:00 PM** period represents the busiest lunch hours.
- A second sales peak occurs between **5:00 PM and 6:00 PM**, indicating strong evening demand.
- Orders begin declining after **8:00 PM**.
- Very few orders are placed after **10:00 PM**.

Business Recommendation

Increase staffing and ingredient preparation during lunch and evening peak hours to improve operational efficiency.

Daily Sales Trend

Insights

- Friday records the highest number of customer orders.
- Thursday and Saturday also perform exceptionally well.
- Weekend demand is consistently higher than weekdays.

Business Recommendation

Maintain additional inventory and staff availability during weekends to meet customer demand.

Monthly Sales Trend

Insights

- July records the highest number of customer orders.
- January is the second-best performing month.
- September and October experience relatively lower demand.

Business Recommendation

Launch promotional campaigns during lower-performing months to improve sales consistency.

Sales Performance

Pizza Category

Insights

- Classic Pizza contributes the highest percentage of total revenue.
- Classic pizzas also account for the highest quantity sold.
- Supreme, Chicken, and Veggie categories contribute similar revenue shares.

Recommendation

Continue promoting the Classic category while introducing marketing campaigns for other categories.

Pizza Size

Insights

- Large pizzas generate the highest percentage of total revenue.
- Medium pizzas contribute the second-highest revenue.
- XL and XXL pizzas contribute only a small percentage of sales.

Recommendation

Focus promotional offers on Large pizzas while evaluating the pricing strategy of XL and XXL sizes.

Best Sellers

Highest Revenue

The Thai Chicken Pizza

- Highest revenue-generating product.
 - Strong customer demand.
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Highest Quantity Sold

The Classic Deluxe Pizza

- Highest quantity sold.
 - Most frequently purchased pizza.
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Highest Number of Orders

The Classic Deluxe Pizza

- Appears in the largest number of customer orders.
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Recommendation

- Feature these pizzas in marketing campaigns.
 - Maintain sufficient ingredient inventory.
 - Create combo offers with beverages and side dishes.
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Worst Sellers

Lowest Revenue

The Brie Carre Pizza

Lowest Quantity Sold

The Brie Carre Pizza

Lowest Number of Orders

The Brie Carre Pizza

Recommendation

- Review pricing.
 - Improve recipe or presentation.
 - Introduce promotional discounts.
 - Evaluate discontinuation if demand remains consistently low.
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Key Business Findings

- The business generated **817.86K** in total revenue.
- Average customer spending is **38.31** per order.
- Customers purchase approximately **2.32 pizzas per order**.

- Friday is the busiest sales day.
 - Lunch and evening hours generate the highest order volumes.
 - July records the highest monthly sales activity.
 - Classic pizzas dominate overall sales performance.
 - Large pizzas contribute the highest share of revenue.
 - The Thai Chicken Pizza is the highest revenue-generating product.
 - The Classic Deluxe Pizza is the best-selling pizza by quantity and orders.
 - The Brie Carre Pizza is the lowest-performing product.
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Dashboard Limitations

- Dataset covers January 2015 to December 2015 only.
 - Customer demographic information is unavailable.
 - Profit analysis cannot be performed because cost data is not provided.
 - Regional or store-level analysis is not available.
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Future Enhancements

Future improvements could include:

- Customer Segmentation
 - Profitability Analysis
 - Inventory Analysis
 - Delivery Performance Dashboard
 - Sales Forecasting
 - Regional Sales Dashboard
 - Customer Lifetime Value Analysis
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Conclusion

This Power BI dashboard transforms raw transactional data into meaningful business insights through data modeling, Power Query transformations, DAX calculations, and interactive visualizations.

The dashboard enables stakeholders to monitor key performance indicators, identify customer purchasing patterns, evaluate product performance, and make data-driven business decisions.

This project demonstrates practical skills in SQL, Power Query, DAX, and Power BI that are commonly required for Data Analyst roles.